

⇒ Algorithm Topos($V[0..n-1][0..n-1], n$)

// purpose: to compute topological order of DAG

// input: adjacency matrix $V[i][j]$ of a dependency graph

// output: topological order $TP[i]$ / "no sequence" if cyclic

compute indegree $[i]$ for all i in $[0, n-1]$

for $i \leftarrow 0$ to $n-1$

indegree $[i] \leftarrow 0$

for $j \leftarrow 0$ to $n-1$

if $V[j][i] = 1$ then

indegree $[i] \leftarrow \text{indegree}[i] + 1$

initialise visited $[i] \leftarrow 0$ for all i in $[0, n-1]$

initialise $TP[i] \leftarrow \text{empty}$, $tp\text{-count} \leftarrow 0$

while true do

$w \leftarrow -1$

for $i \leftarrow 0$ to $n-1$ do

if visited $[i] = 0$ & indegree $[i] = 0$ then

$w \leftarrow i$

break

if $w = -1$ then break - no eligible vertex found

$TP[tp\text{-count}] \leftarrow w$

$tp\text{-count} \leftarrow tp\text{-count} + 1$

visited $[w] \leftarrow 1$

for $i \leftarrow 0$ to $n-1$ do - decrement dependents of w

if $V[w][i] = 1$ then

$\text{indegree}[i] \leftarrow \text{indegree}[i] - 1$

end while

if $\text{tp-count} < n$ then

 return "No sequence (cycle detected)"

else

 return $\text{TP}[0, \text{tp-count} - 1]$

~~Boyle~~
13/11/26.